

common manuals

MkXL manuals

MkIV manuals

MkII manuals

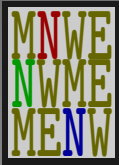
The manuals

Welcome to the suite of ConT_EXt manuals. These manuals not only cover the macro package itself, but also the tools that come with it. In this suite you will also find manuals on how to use ConT_EXt for processing xml. Fonts and MetaPost graphic are discussed in dedicated manuals.

On the following pages, the main manuals are shown large, while their screen companions are shown in the bottom right corner of a page. Clicking on a picture brings you to the document. Some manuals come in more than one language, in which case small pictures of the title pages are shown.

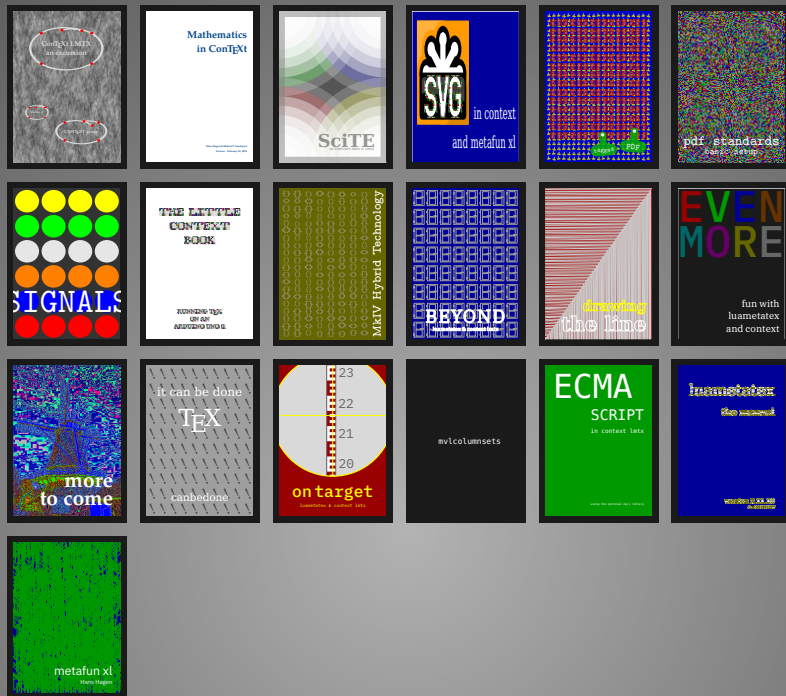
The next pages show overviews of manuals that are specific for MkII, MkIV and , as well as obsolete manuals. The archived manuals cover MkII but can still be of interest, if only because they give an idea about how all evolved.

This is a snapshot. Check the distribution for the latest versions and extra manuals.



Common manuals

These are generic manuals. The more version specific manuals are collected on the following pages, including some historic ones. In addition to the manuals you can have a look at the Wiki and some manuals (in various languages) by users that can be found on the Internet.



MkXL manuals

All the good things come together in the LMTX where we pair LuaMetaTeX with MkXL. This combination is more powerful and performs better. Most of the MkIV manuals apply but there are some that are more exclusive to the latest greatest. Manuals tagged at MkIV might not compile with LuaMetaTeX due to new features but still be usefull for that version.



MkIV manuals

Here you will find the manuals that describe MkIV functionality and/or features not present in MkII. But again we froze a version when we moved on, this time to LuaMeta $\TeX$. The MkIV version is the test-case for Lua $\TeX$ so it will adapt when needed.



MkIV manuals

Here you will find the manuals that describe MkIV functionality and/or features not present in MkII. But again we froze a version when we moved on, this time to LuaMetaTeX. The MkIV version is the test-case for LuaTeX so it will adapt when needed.

(continued)



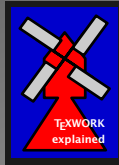
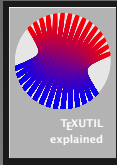


MkII manuals

Although MkII and MkIV are rather compatible, there are some differences. Also, as MkII is frozen there are more features MkIV. Also because we moved to Unicode and OpenType there are differences in those areas.

When you want to process these manuals you need a MkII installation which includes fonts no longer present in the later versions. We will not update these manuals! They describe history.





Obsolete manuals

We keep some of the old manuals around for historic reasons. Some of what is described might still float around in the distribution but is likely replaced by more modern and hip variants.





Read Me First

Read Me

It's in the name: you should read this file. Not so much because the content should bother you, but more because it gives you an idea about what we have in mind with making ConT_EXt available for everyone.

ConT_EXt is completely free software, which does not mean that there are no restrictions on redistributing and changing the files. When you want to redistribute (changed) source code, please read this licence first.



ConTeXt

commands

EN

Quick References

This quick reference manual does not replace the other manuals, but advanced users can benefit from its compactness. The manual can be generated on the user's system, since the style and database that is needed is part of the distribution.



setup.cz





A 3x4 grid of letters. The letters are arranged as follows:

M	N	W	E
N	W	M	E
M	E	N	W

The letters are colored as follows:

- Row 1: M (olive), N (red), W (olive), E (olive)
- Row 2: N (green), W (olive), M (olive), E (olive)
- Row 3: M (olive), E (olive), N (blue), W (olive)

MWE

When you run into troubles the best place to get help is the mailing list. There the magic word is .

What is ConT_EXt

This is a one page summary of what ConT_EXt is about.

There are several ways to deploy ConT_EXt. Most common is to let it render a document, in which case you install it on a system and use an editor to input your document and from a console or by clicking some key trigger a run. The input can be structured using T_EX macros but it can also be xml, or some other format that gets converted to ConT_EXt commands before processing.

Instead you can also use ConT_EXt as a more hidden application, for instance in a web service or rendering component in a larger application. In that case the end user is not really aware that T_EX is being used.

No matter how you use ConT_EXt, you need to install it first. You can for instance use T_EXlive or another distribution to pick up ConT_EXt, but you can also install it using the archive (snapshot), in which case you also need to pick up the engine (for instance LuaT_EX) and a basic set of fonts. In order to make installation easy we provide a so called standalone distribution that has all you need.

The standalone ConT_EXt distribution has the following characteristics:

- The installation is self contained. Apart from resources like fonts, the T_EX macros, Lua code and MetaPost helpers are provided in one package.
- There is only one binary involved: LuaMetaT_EX. The source code of this program is integral part of the ConT_EXt distribution (per end 2019). A user should be able to compile the program if needed. There is no dependency on additional libraries other than those that make up the operating system.
- The core system is able to typeset documents in an efficient way. The memory footprint is decent and performance acceptable, also on low power devices and virtual machines. We try not to provide bloatware.
- The official user interface is stable and the implementation of core components will not change fundamentally. When something can be improved it will be. One can use a snapshot for long term stability.
- Support is provided by means of a mailing list, a wiki, documentation, meetings, etc. If needed you can consult (or hire) support. There are enough experienced users out there to get you going.

The first version of ConT_EXt, now tagged MkII, has been around since 1995 and (still) runs on top of pdfT_EX. The development of its successor MkIV started in 2005 as part of the LuaT_EX development and still carries on. The most recent incarnation is lmtx, which is MkIV but tuned for LuaMetaT_EX, the lean and mean successor of LuaT_EX.

ConT_EXt lmtx — factsheet

What is LuaMetaTeX

This is a one page summary of what Lua-MetaTeX is about.

The LuaMetaTeX engine is a follow up on LuaTeX. It integrates the TeX text rendering engine, the MetaPost graphic engine and the Lua script interpreter. The development is part of the ConTeXt macro package development. This macro package tightly integrates the three subsystems. The LuaMetaTeX code is part of the ConTeXt distribution.

The LuaMetaTeX engine is lean and mean. There is for instance no backend code present. In ConTeXt this is handled in Lua. Graphic inclusion is also delegated to Lua. The TeX frontend is a slightly extended version of the LuaTeX one. System dependencies are minimized. Where possible we stay close to the original TeX concept because that is a well documented reference. The binary can also be used as stand alone Lua engine.

The MetaPost library also has access to Lua, which means that the language can be enhanced and functionality added on demand. There are several graphic libraries provided in ConTeXt. This graphical language is efficient in runtime and graphical output. In combination with Lua we have a high performance graphical subsystem that can handle a huge amount of data. Additional text (like labels) is typeset at high quality.

The Lua code that comes with ConTeXt contains a lot of helper code which means that one can set up selfcontained workflows without many extra dependencies. Documents can be encoded in TeX, Lua, xml or whatever suits. There is support for databases too.

The ConTeXt code base evolved over time. The basic functionality is quite stable. The move from MkII to MkIV to lmtx has been gradual. The efficiency in terms of code and performance has been improved stepwise. Development continues and beta releases occur on a regular basis. The ConTeXt user community is quite willing to experiment with betas that can be installed alongside stable snapshots.

The installation is relatively small (fonts make up much of it) and updating is easy. We operate in the TeX Directory Structure, which is a proven concept.

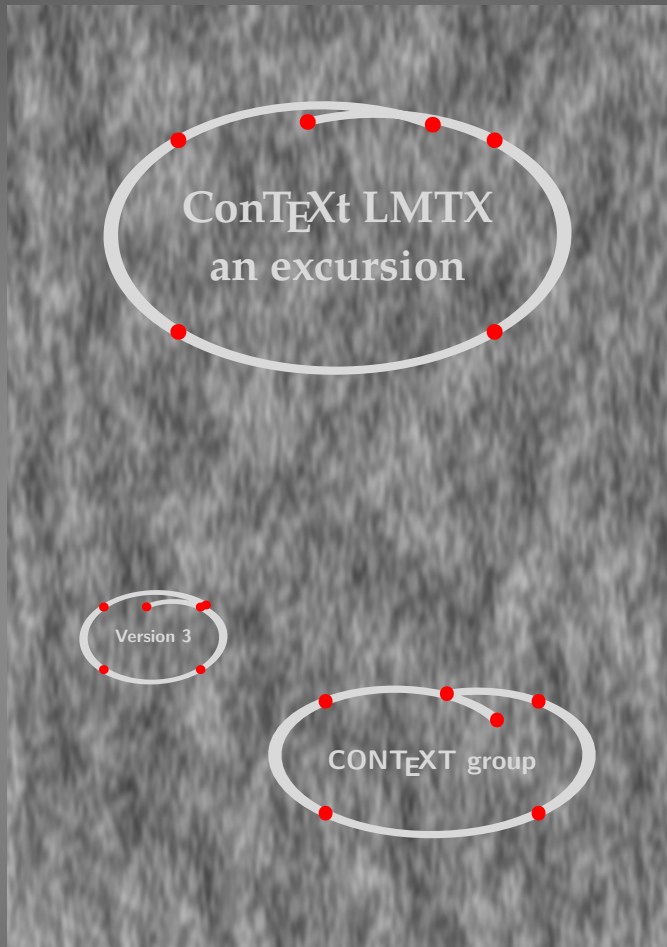
LuaMetaTeX — factsheet

Input methods

These few pages show various input methods in ConT_EXt. There are more so reading might trigger you to look at various manuals, articles and examples.

Excursion

This is the follow up on the getting started manual. It gives a overview of the most important ConT_EXt commands. The older manual covers MkII and MkIV.



Mathematics in ConT_EXt

Hans Hagen & Mikael P. Sundqvist
Version: February 28, 2025

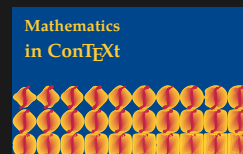
Math In ConT_EXt

This manual describes how to typeset math in ConT_EXt LMTX. It discusses the various mechanisms but also shows how to code without tweaking in order to keep a clean source. It uses the advanced math rendering options available in LuaMetaT_EX and has numerous examples.



examples-

matheanings





SciTE

IN CONTEXT MkIV & LMTX

SciTE in ConT_EXt

SciTE is an editor and these manuals describe how to configure it for use with ConT_EXt and MetaFun. Beware, the mscite manuals are the old ones, still valid for traditional lexing, while the readme version describes the latest greatest lexers.

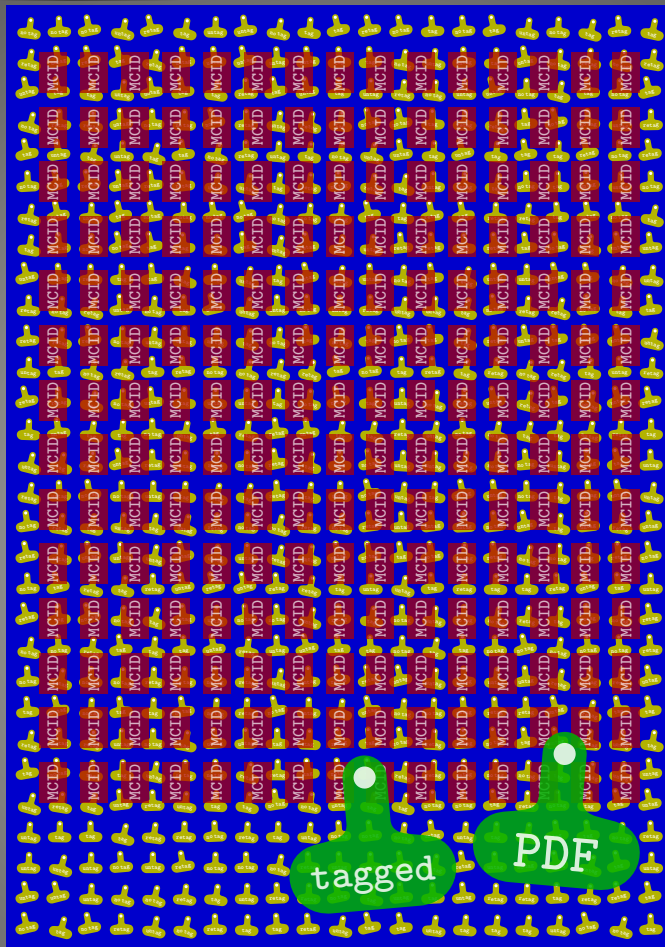
SVG

You can let ConT_EXt convert svg images. It does so with the help of MetaPost.



in context
and metafun xl



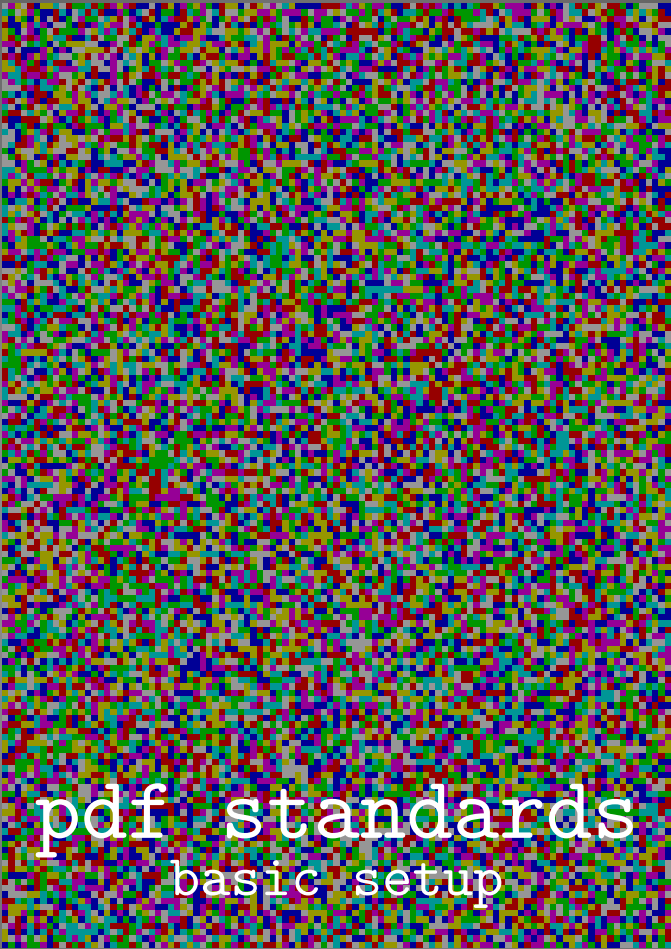


Tagging

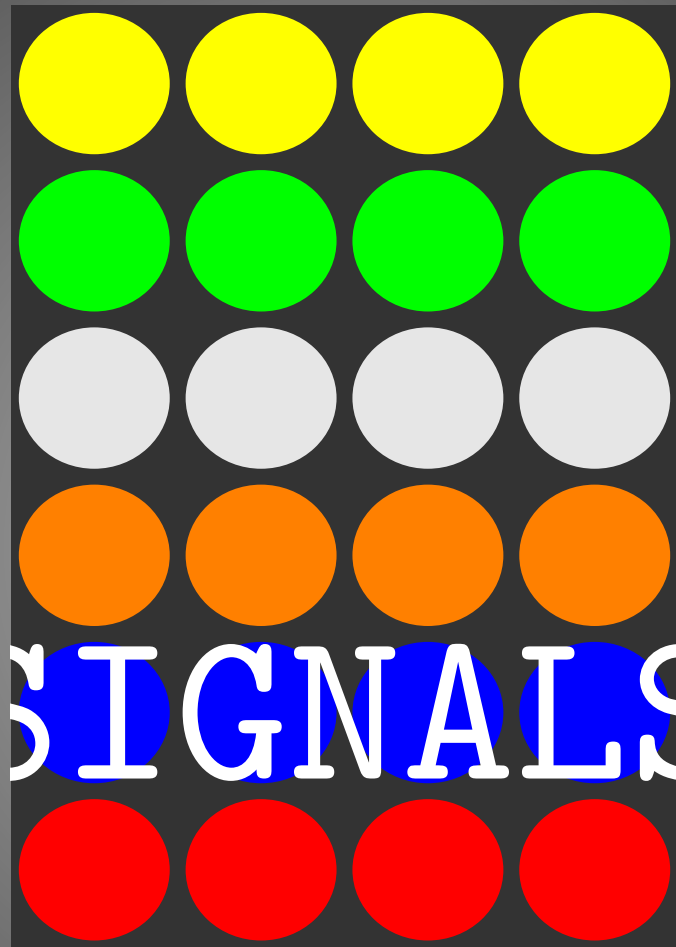
When you are forced to provide tagged pdf
this manual is your friend.

Standards

In case you're forced to use pdfstandards, this manual can be of help.



pdf standards
basic setup



Signals

This manual introduces you to the signal progress and error feature assisted by the 'ConT_EXt watch' presented at the 2025 meeting. We're talking of a gadget here.

THE LITTLE CONTEXT BOOK

RUNNING T_EX
ON AN
AROUND UNO 0

Little

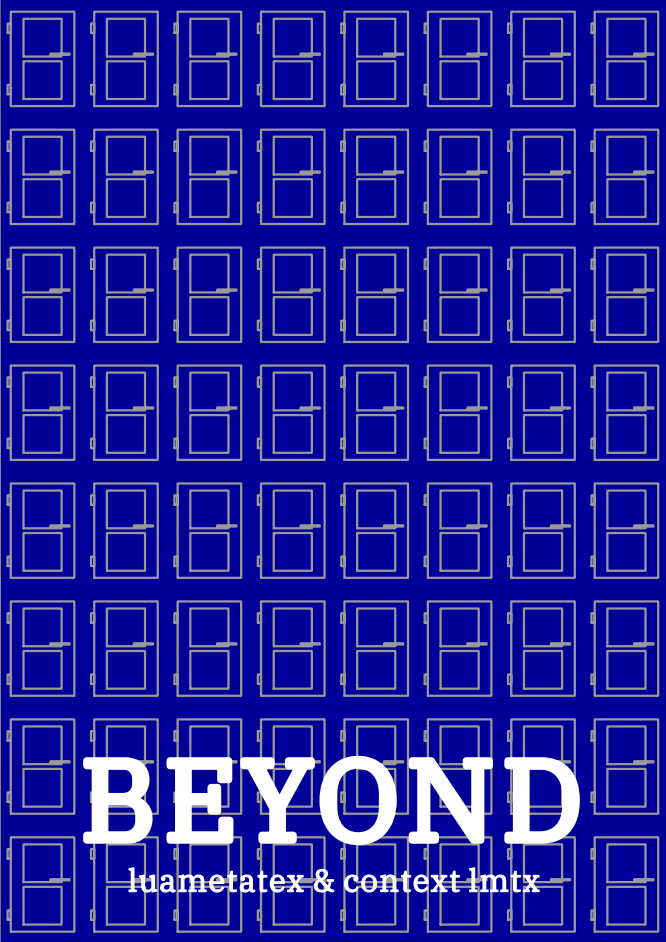
This little manual introduces the ‘little context book’ that we presented at BachoT_EX 2026 as statement against the energy and environmental side effects of artificial intelligence (datacenters).

MkXL development: hybrid

This document is one in the series that keeps track of various developments in Lua-TeX, LuaMetaTeX, ConTeXt and LuaMetaFun.

Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.





BEYOND

luametatex & context lmtx

MkXL development: beyond

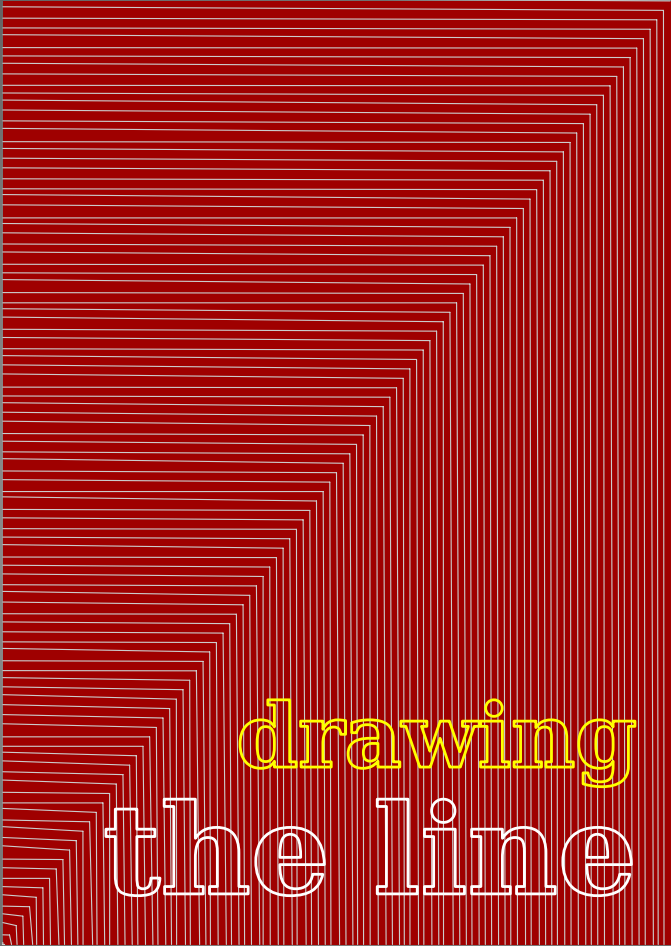
This document is one in the series that keeps track of various developments in Lua_T_EX , LuaMeta_T_EX, Con_T_EXt and LuaMetaFun.

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MkXL development: drawing lines

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drawing
the line



EVEN
MORE

fun with
luametatex
and context

MkXL development: even more

This document is one in the series that keeps track of various developments in Lua_T_EX , LuaMeta_T_EX, Con_T_EXt and LuaMetaFun.

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MkXL development: more to come

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it can be done

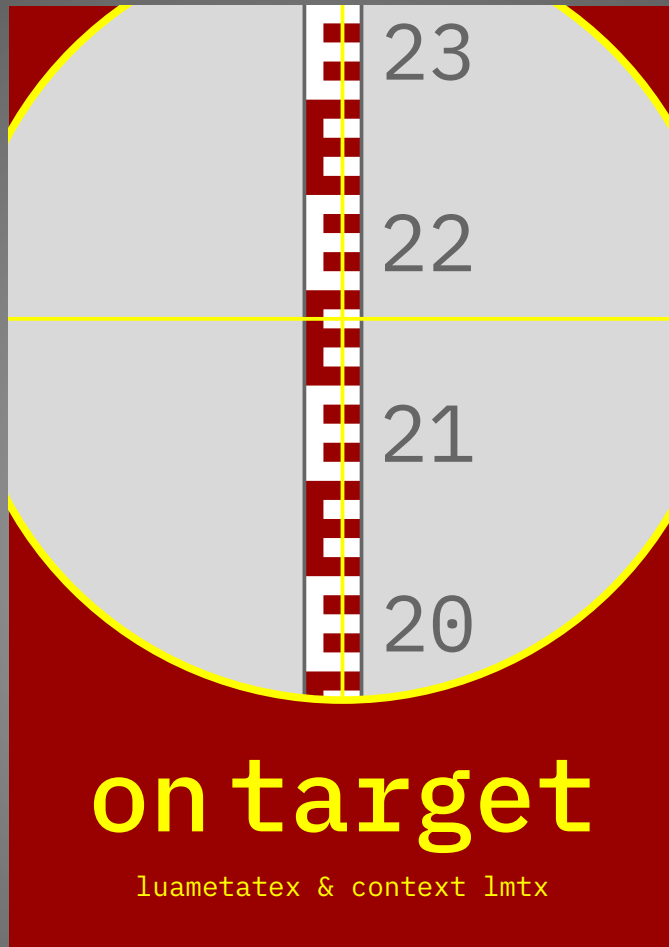
TeX

canbedone

MkXL development: can be done

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MkXL development: on target

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Here you can also find explanations not found in manuals. As with earlier development stories, some were written as articles for user group journals.

Columnsets

The new columnset implementation does whole-set optimization, is more robust and efficient and also more powerful. This manual explains how to define and populate pages and create parallel versions of documents. This is work in progress.

```
name: documents/general/manuals/mkx1/mv1columnsets.pdf
file: documents/general/manuals/mkx1/mv1columnsets.pdf
state: unknown
```

ECMA SCRIPT

in context lmtx

using the optional mujs library

ECMAScript in Con- Text LMTX

It is possible to embed ecmascript (of which JavaScript is an example) in ConT_EXt LMTX. The (external) library used is .

luametatex

the manual

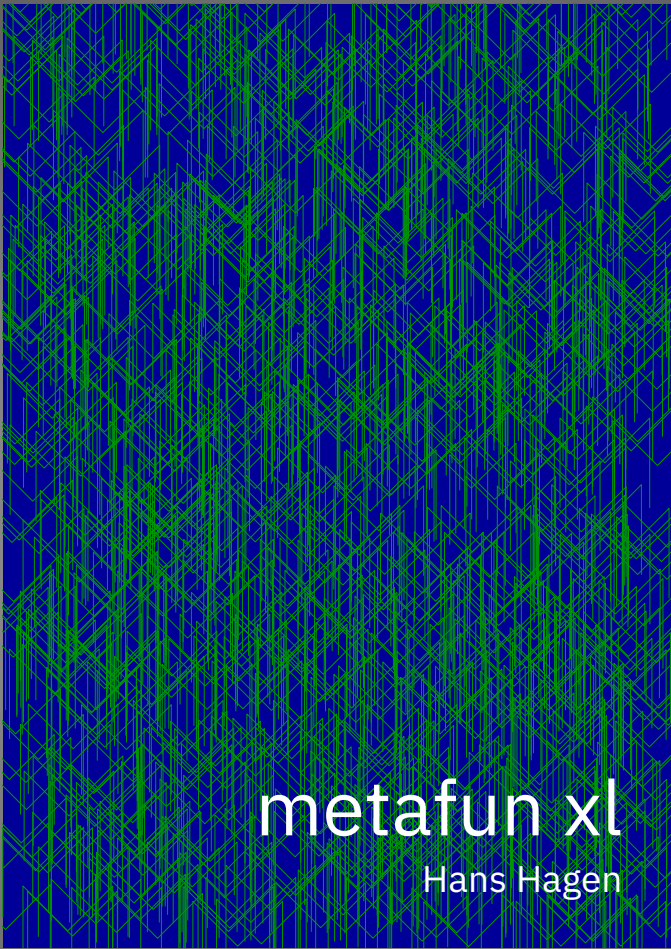
version 2.11.10

dev id 20260917

LuaMetaTeX

The LMTX version of ConT_EXt uses the LuaMetaT_EX engine. This engine is a follow up on LuaT_EX and is an integral part of ConT_EXt development. This manual describes the current (official) features.

The related manual about primitives discusses some of the new primitives that relate to low level macro programming.



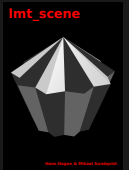
metafun xl

Hans Hagen

LuaMetaFun

In LuaMetaTeX we use an enhanced version of MetaPost that is supported by extensions to MetaFun discussed in this manual.

There is a dedicated manual for 3D graphics that describes in great detail how to generate implicit, parametric, and related surface plots in high quality bitmap format bitmaps.





ConTEXt Mark IV an excursion



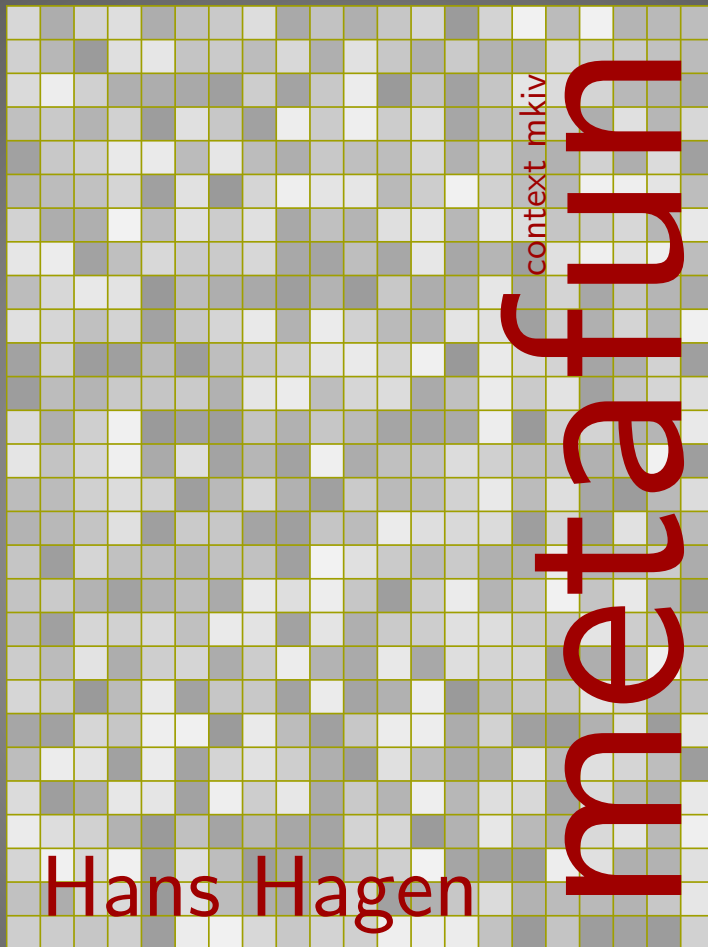
English



Ton Otten
PRAGMA ADE

Getting started

Although meant for beginners, these manuals shows already a lot of what ConTEXt can do for you. They also demonstrate that TEX documents can be colorful and can contain lots of graphics. See the excursion manual for an update!



MetaFun

If you like graphics, you may like MetaFun, a collection of MetaPost macros. The manual covers most of MetaPost, as well as the interface between this graphical environment and ConT_EXt. There are numerous examples, that give you an impression about the power of this graphical system as well as the strength of the combination with T_EX.





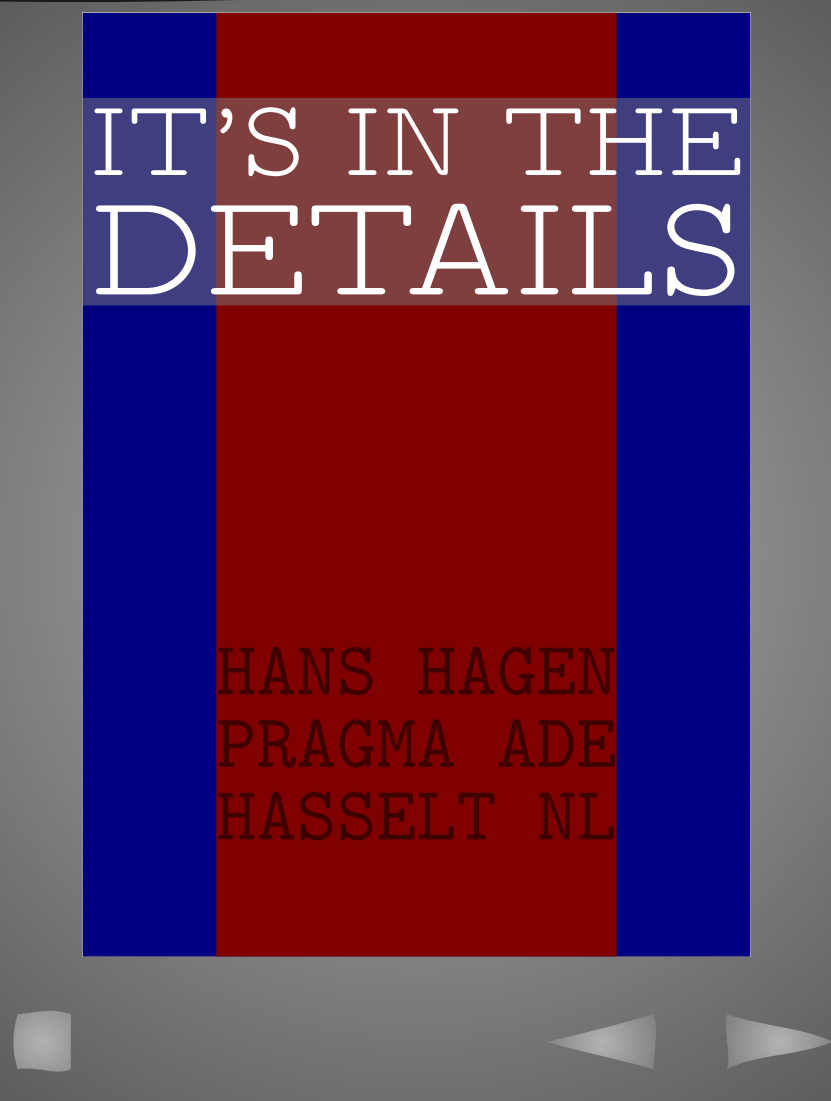
ConTeXt

Hans Hagen

Interaction

Producing interactive documents have always been an integral part of ConTeXt. This manual describes how to configure hyperlinks, comments, attachments, forms and also how to add navigational elements to a document designed for display.





IT'S IN THE DETAILS

HANS HAGEN
PRAGMA ADE
HASSELT NL

It's in the details

This manual is meant for users who want to divert from the more or less traditional looking T_EX documents. There is a strong focus on elements that determine the look and feel of a document, like graphics. (This manual is unfinished)



ConT_EXt MkII – MkIV, the history of LuaT_EX

This document keeps track of the development history of both ConT_EXt (mkiv) as well as LuaT_EX. It is also one of our torture tests for both (rather related) systems.



MkIV Hybrid Technology

MkIV hybrid technology

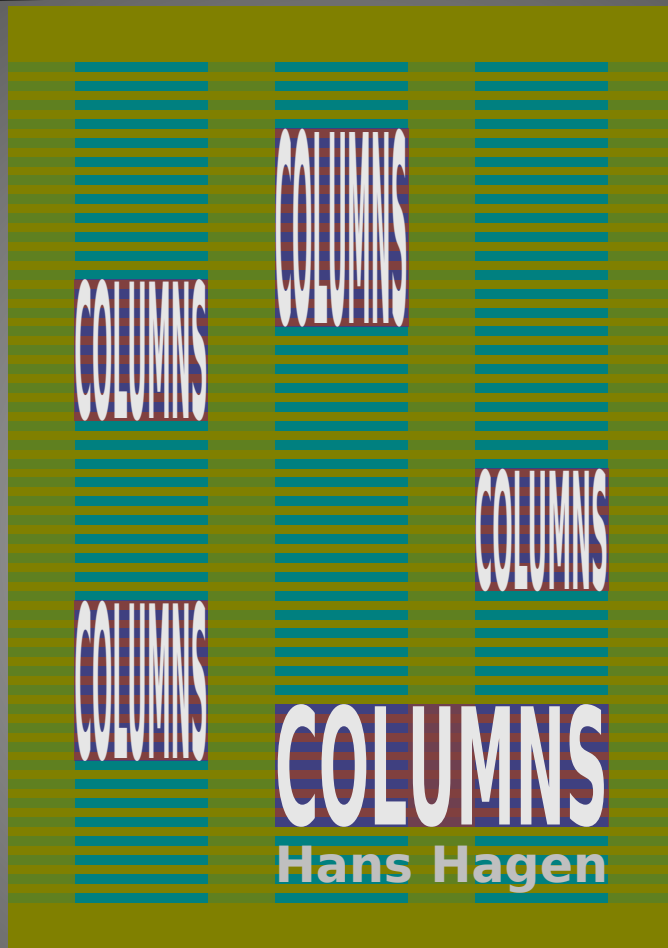
This document keeps track of the development history of both ConT_EXt (mkiv) as well as LuaT_EX from the moment we considered ourselves to be halfway in the project. Like the MK document it is also one of our torture tests.

Many of the chapters of MK were first published as articles and the same is true for this document. So, the version published on the web lags behind as we don't want to compete with the user group journals.

LIBRARIES

In LuaT_EX we use the ffi interface to implement support for external (binary) libraries. In LuaMetaT_EX we use so called optional libraries instead. This manual explains this in more detail.

In LuaT_EX we use the ffi interface to implement support for external (binary) libraries. In LuaMetaT_EX we use so called optional libraries instead. This manual explains this in more detail.



Columnsets (old)

Column sets can be used for quite complex but nice looking layouts. They are used for magazine like layouts and mix well with explicitly placed graphics. The MkIV version is a bit different from the MkII version but uses the same principles.



Math

This a preliminary manual about some aspects of math typesetting. It is not a replacement for the Knuthian references.

Spacing

This a work-in-progress manual about aspects of spacing in ConT_EXt MkIV.

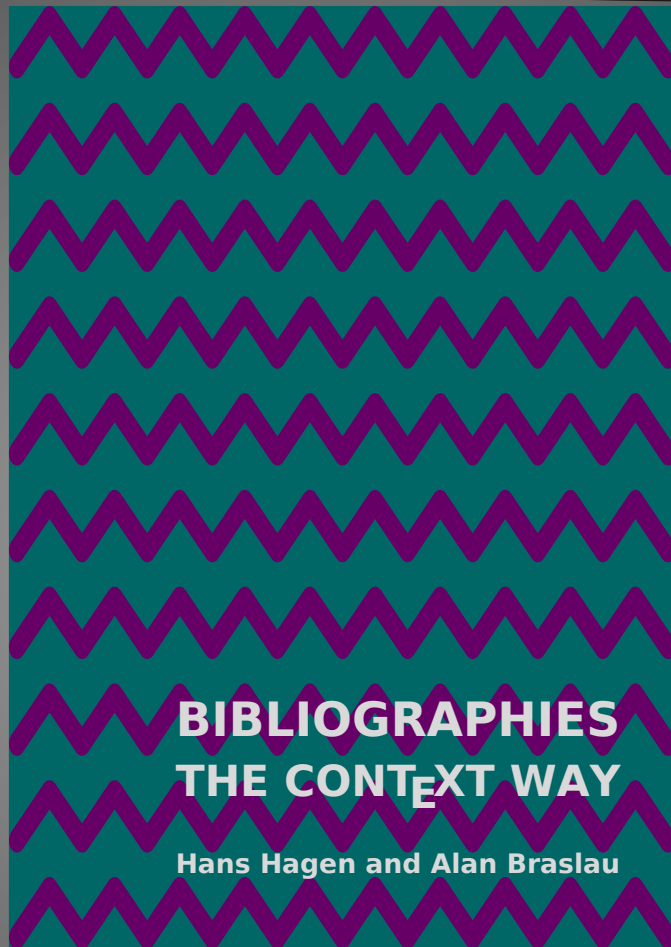
Spacing in ConT_EXt



workflow
support in context

Workflows

The ConT_EXt ecosystem is of course centered around typesetting but in addition comes with all kind of tools and subsystems for managing the process. Here we collect some tips.



Publications

In ConT_EXt we support the bibT_EX format for handling references. The subsystem for dealing is flexible enough to deal with many situations and is extensible as well. It does not depend on external tools and is driven by Lua on the one hand and ConT_EXt setups on the other.



low level

TeX

lowlevel

Lowlevel

This is a series of small manuals explaining subsystems of ConT_EXt often in the perspective of LuaMetaT_EX. Some are real lowlevel.

Languages

The ability to deal with many languages is an important property of T_EX systems. Here we cover aspects like hyphenation and language dependent labels.



Languages in Context
explaining luatex and mkiv

Hans Hagen
PRAGMA ADE

Coloring ConTeXt
explaining luatex and mkiv

Hans Hagen
PRAGMA ADE

Colors

Color support is like fonts and languages a core mechanism. This manual is part of the more technical description of features like that.

hans hagen

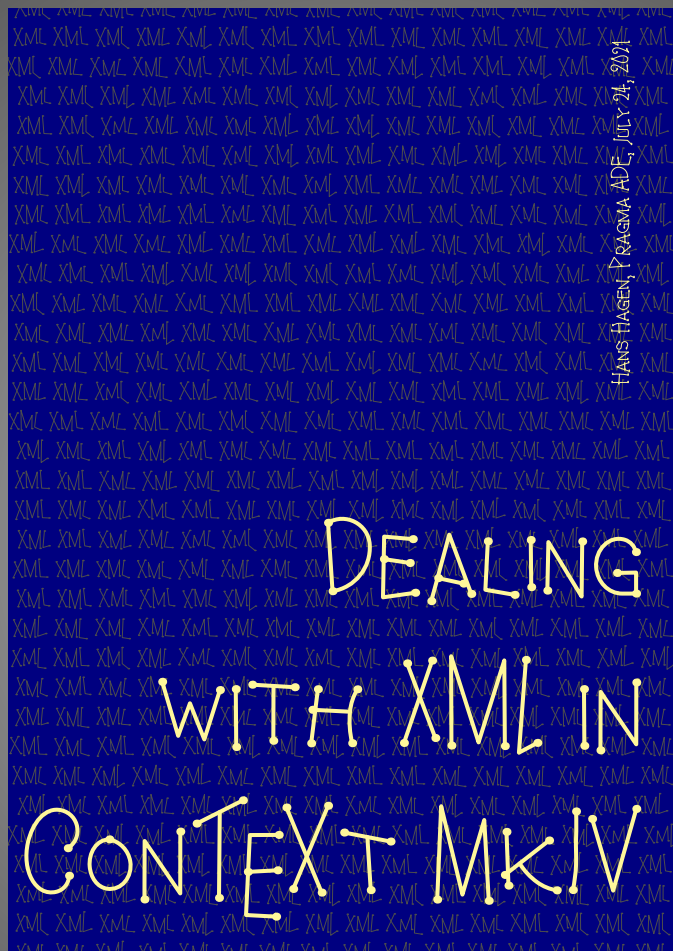
about

luatex and context

About LuaT_EX and ConT_EXt

This is the third document in the series about the development of LuaT_EX and MkIV. This one goes under the name ‘about’ as one might wonder what all this development is about. After all we’ve now reached a state where we can think about future applications instead of improving older features as that process is ongoing.

As we’re a bit beyond experimenting now, the focus will be on practical usage and of course we target on applications that the Lua and T_EX combination makes possible, either new or in a renewed form.



Dealing with XML

This manual explains how to define styles for tree based processing of xml files. This variant showed up in MkIV. The manual also contains examples of filtering content.

ConTeXt Lua Documents

Hans Hagen

ConTeXt Lua Documents

This manual describes how to generate documents (structure as well as content) using Lua exclusively. Of course you can also embed such code in your normal $\TeX$ documents but using Lua has some advantages when you deal with for instance database output.



luatools mtxrun context

Luatools, Mtxrun & Context

Here we discuss the main tools on the Con-
T_EXt suite of programs. We focus on the lua-
tools tree handler, the mtxrun script man-
ager and the process management tool Con-
T_EXt.



HANS HAGEN
CONTEXT MKIV

EXTREME TABLES

Extreme Tables

This is a short introduction to yet another table mechanism built in ConT_EXt. It is a variant of the so called natural tables but it has a different configuration. These tables are faster to process and can span lots of pages.

HANS HAGEN
CONTEXT MKIV

S T I N G

Units

As part of physics support the ConT_EXt core provides a mechanism for typesetting units. This manual describes the basics as well as explains how additional units can be added and extra variants of the command can be configured. The manual also introduces the related digits mechanism. The built in support for units should not be confused with the older (incompatible but conceptually similar) units module.

A01	B01	C01	D01	E01	F01	G01	H01	I01	J01	K01	L01
A02	B02	C02	D02	E02	F02	G02	H02	I02	J02	K02	L02
A03	B03	C03	D03	E03	F03	G03	H03	I03	J03	K03	L03
A04	B04	C04	D04	E04	F04	G04	H04	I04	J04	K04	L04
A05	B05	C05	D05	E05	F05	G05	H05	I05	J05	K05	L05
A06	B06	C06	D06	E06	F06	G06	H06	I06	J06	K06	L06
A07	B07	C07	D07	E07	F07	G07	H07	I07	J07	K07	L07
A08	B08	C08	D08	E08	F08	G08	H08	I08	J08	K08	L08
A09	B09	C09	D09	E09	F09	G09	H09	I09	J09	K09	L09
A10	B10	C10	D10	E10	F10	G10	H10	I10	J10	K10	L10
A11	B11	C11	D11	E11	F11	G11	H11	I11	J11	K11	L11
A12	B12	C12	D12	E12	F12	G12	H12	I12	J12	K12	L12
A13	B13	C13	D13	E13	F13	G13	H13	I13	J13	K13	L13
A14	B14	C14	D14	E14	F14	G14	H14	I14	J14	K14	L14
A15	B15	C15	D15	E15	F15	G15	H15	I15	J15	K15	L15
A16	B16	C16	D16	E16	F16	G16	H16	I16	J16	K16	L16

ConTeXt MkIV

Hans Hagen

May 12, 2016

Simple Spreadsheets

Simple Spreadsheets

This module provides an easy way to add calculations to a document in a tabular form. It is not a replacement for a decent spreadsheet program but fits well into regular document processing as done by ConTeXt.



LMX templates

Templates as described here can be used to construct ConT_EXt documents using a more programmatic approach. The method discussed will stay but might get extended. This mechanism also introduces two new dialects: MkIX and MkXI.



Lua

the context libraries

Lua libraries

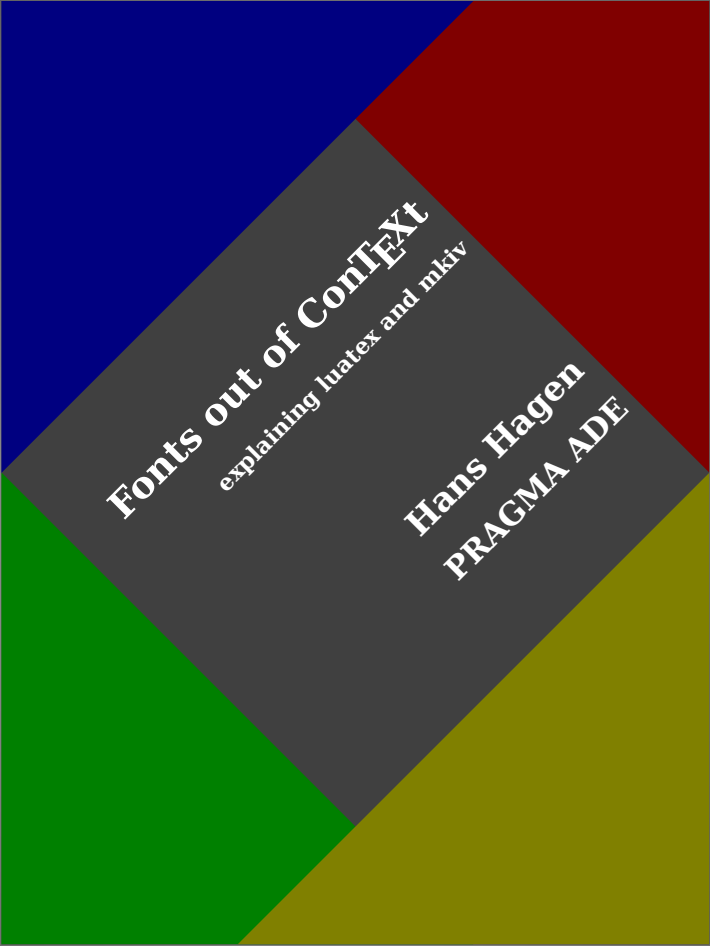
This manual describes how to use generic Lua modules outside ConT_EXt. The helper functions themselves are discussed in the cld manuals.



SQL CONTEXT

SQL in context

The ConT_EXt infrastructure can be quite handy to process sql output. This manual describes how integrate mysql support into your styles. The libraries can also be used independent from ConT_EXt but fit into the package.



Fonts out of Context

explaining luatex and mkiv

Hans Hagen
PRAGMA ADE

Fonts out of context

In $\text{T}_{\text{E}}\text{X}$ and therefore in $\text{ConT}_{\text{E}}\text{Xt}$ fonts play an important role. This document describes some of the characteristics of the MkIV font system. It is not a manual about using fonts, although some details can be found only here.



Math

This manual describes a few aspects of typesetting mathematics in ConT_EXt and will evolve over time.

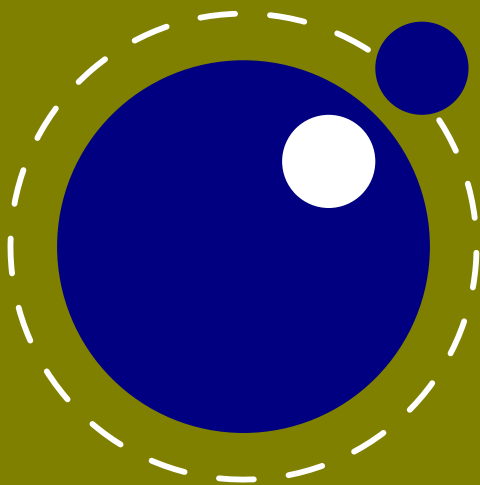
<div />
<div >
</div>

**exporting
xml and epub
from context**

Epub

The export option in ConT_EXt can produce a basic set of xml and epub files that can be either used directly (using a css) or enhanced for usage otherwise. This manual gives an overview of the process.

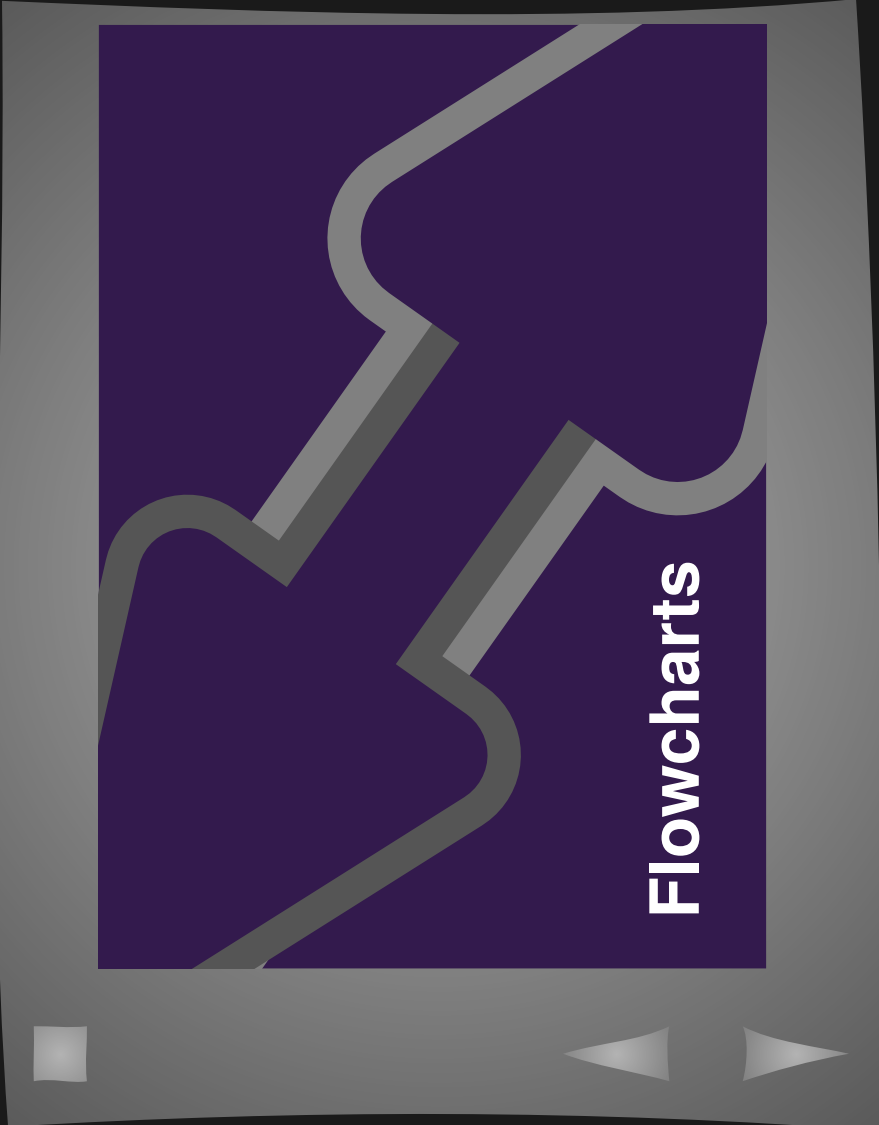
LuaT_EX Reference Manual



stable
July 2021
Version 1.10

LuaT_EX

The MkIV version of ConT_EXt uses the LuaT_EX engine. This engine is an ongoing development and happens in the scope of ConT_EXt development. This manual describes the current version and is offered here for convenience.



Flowcharts

Flowcharts

The flowchart module is an old one that has been around for a while. It got updated to MkIV and wil stay around.

Hans Magen

This is yet another document in the series about the development of LuaT_EX and MkIV. This time we use the tag ‘onandon’ to indicate that we never seem to stop moving forwards. The content lags a bit behind in the sense that some chapters are first published in user group journals.





ConTEXt
lmtx

FOLLOWING UP

Followingup

This is the fifth document in the series about the development of LuaTEX and MkIV, but here we focus on LuaMetaTEX instead. It is a relative small set of progress reports that describes the steps to get there. This engine is a follow up on LuaTEX, the engine that is used for the ConTEXt version tagged as LMTX. While the functionality of its ancestor is more or less set in stone, so that it can be used in other macro packages, this follow up permits further experiments in ConTEXt.

RULES

HANS HAGEN
A CONTEXT MKIV MANUAL

Rules

In this manual we cover some aspects of drawing (ornamental) rules in ConT_EXt using native rule operators as well as MetaPost.

l2r

r2l

a few tips

hans hagen

Bidi

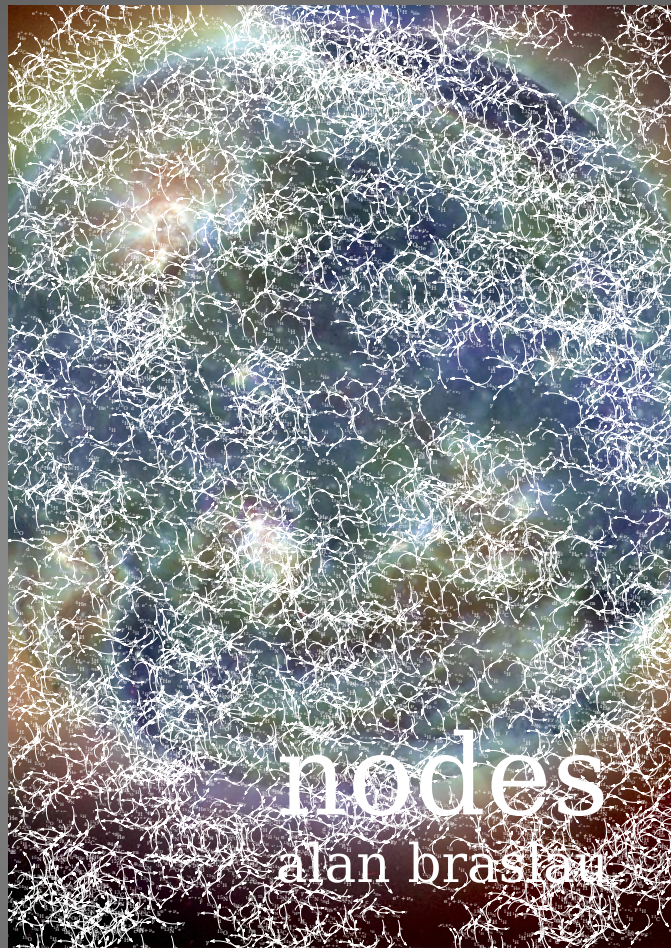
Right to left typesetting involves directives, fonts, heuristics, and a sometimes dedicated layout. In ConT_EXt some mechanism are direction aware. Here we discuss some details.

Context^t Musings

hanS hagen

Musings

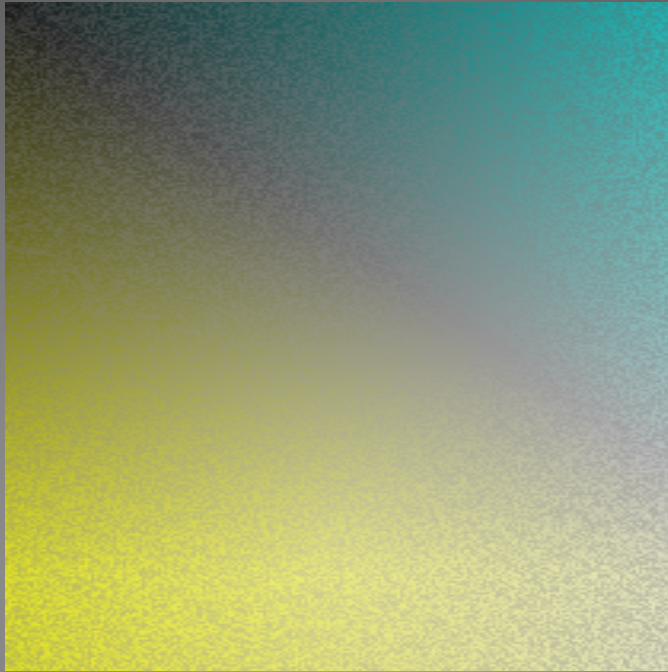
In this manual we collect articles that don't fit into another manual or collection. Some relate to talks, other to experiences or observations. They are often opinions.



Nodes

This manual is about a rather neat set of macros to produce node related drawings in MetaPost and ConT_EXt like charts and trees. It also presents some tricks that can be applied elsewhere.





Graphics

Hans Hagen

Graphics

This manual explains how to insert images into a document.





PDF
PDF
PDF
PDF
PDF
PDF

merging, embedding, fixing
and messing a bit around
in context lmtx

PDFmerge

This manual explains how to efficiently include pages from pdf files into documents. It also goes into detail about some cleanup and manipulation of the content.

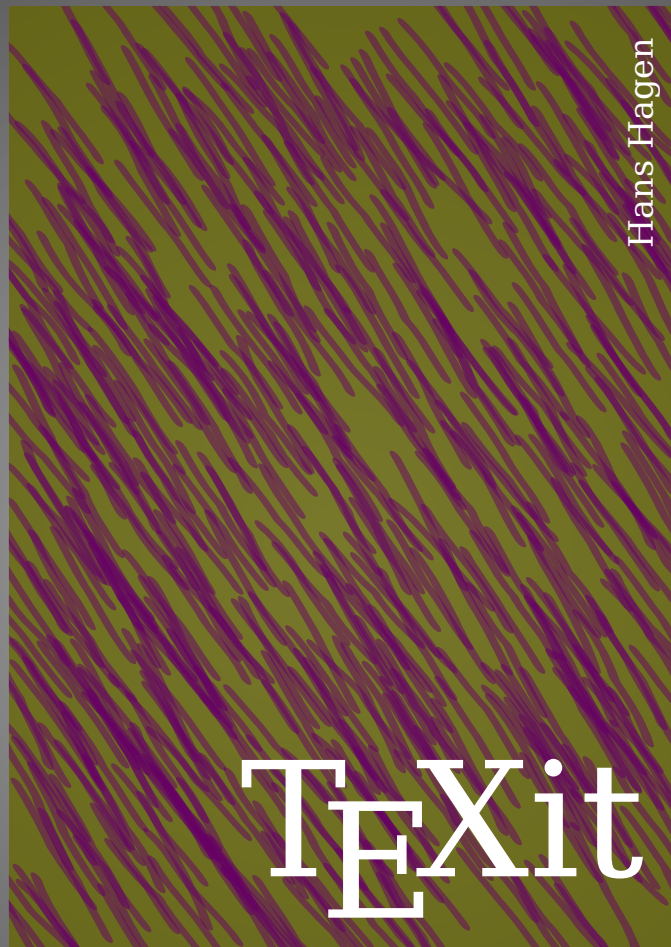


Still going on

Hans Hagen

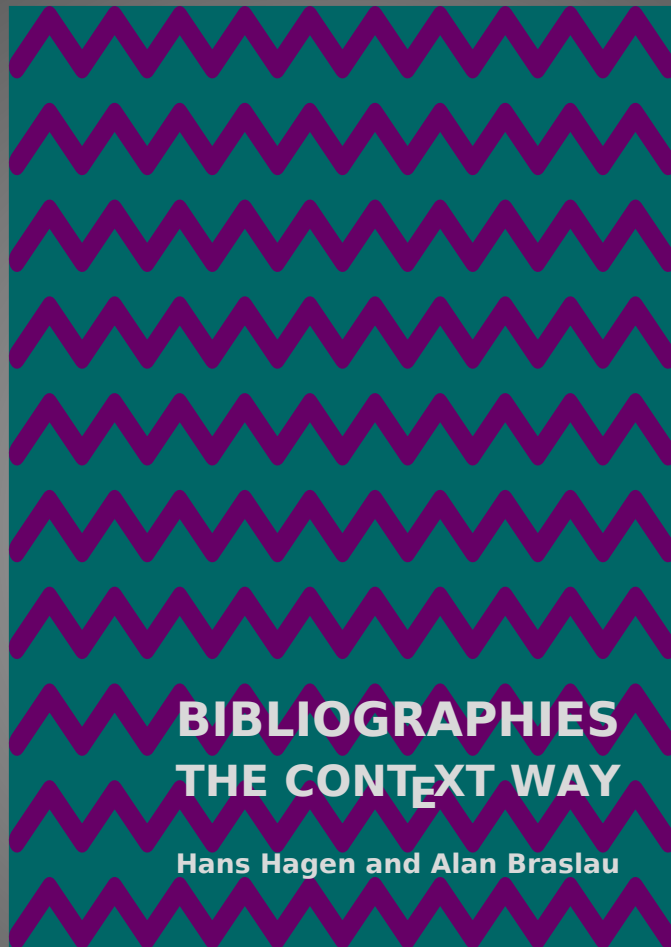
Still

This is the fourth document describing the history of LuaTeX. Most of the development is done, but we keep on playing with the possibilities it offers for ConTeXt. We finally arrive at version 1.0 too.



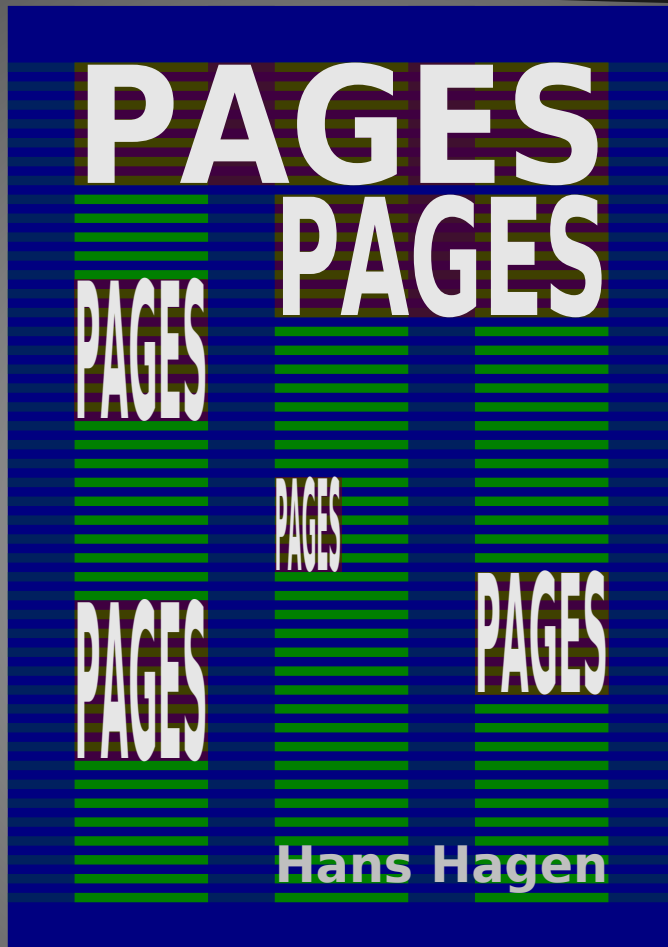
TeXit

Sometimes questions on the mailing list pop up that demand a bit more tex-nical explanation. This manual will collect explanations and insights that don't fit into regular manuals.



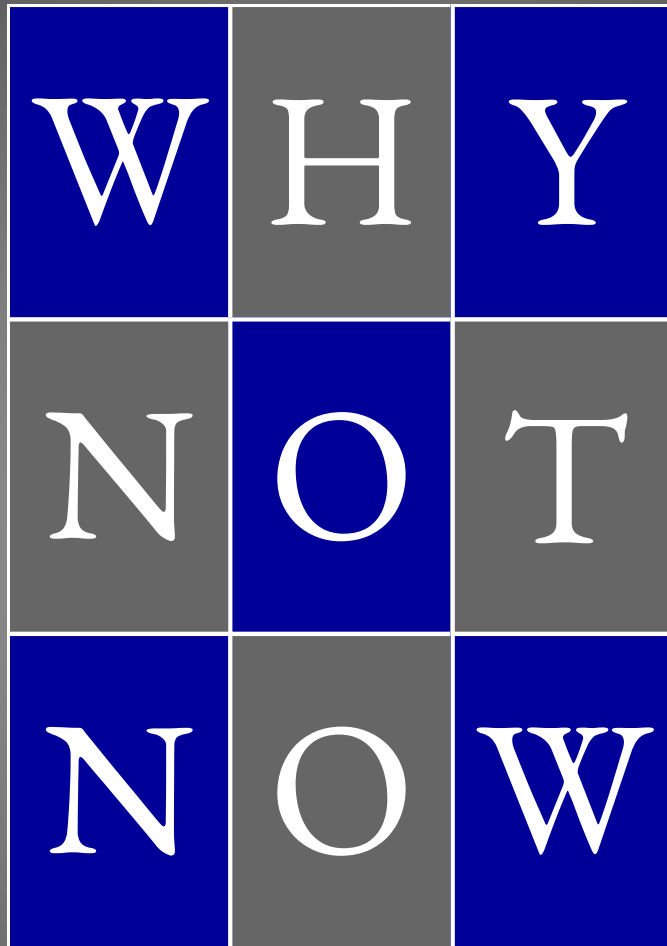
Publications

Like any macro package ConT_EXt has to support bibliographies. This manual describes in great detail how to copye with this, and especially bibT_EX databases and finetuning the rendering. We got rid of dependencies of external programs and all happens in Lua. This also opens up access to the data to users for various purposes.



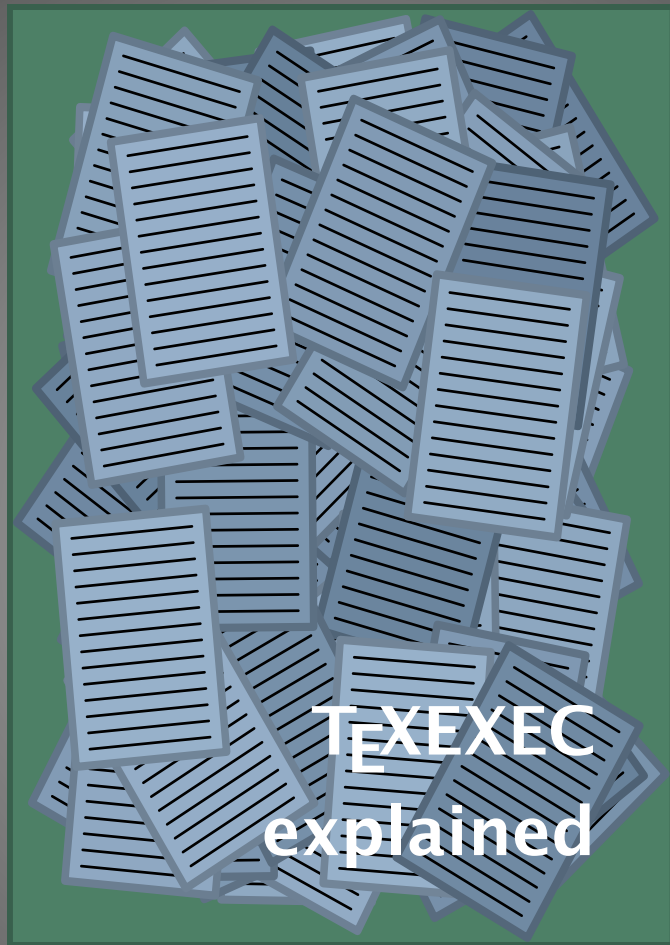
Page columns

There are several column mechanisms and this is one of them. It boils down to treating each column as a page which in turn means that we can do for instance side floats. This manual might also give you an idea about its usability.



Not now

This is more an excuse manual: why are some features not supported or limited.



$\text{T}_{\text{E}}\text{X}$ exec

Traditional $\text{T}_{\text{E}}\text{X}$ is hard to control on the commandline. This is why Con $\text{T}_{\text{E}}\text{X}$ t comes with $\text{T}_{\text{E}}\text{X}$ exec, a Perl script that makes document processing more convenient. This script also helps you to postprocess pdf files, typeset Con $\text{T}_{\text{E}}\text{X}$ t documentation, arrange pages, and manage files.



Typographic Programming

Designing styles is a mixture of making the right decisions in setting up the layout, finding the right values for the parameters that determine the typographic quality of the paragraph and page, and writing programs that take care of constructing the special elements that make up the page. This document tries to provide some insight in these matters and will be completed when we have time or reason.



Typographic
Programming

Hans Hagen

Fonts in ConT_EXt

Hans Hagen
October 2001

Fonts

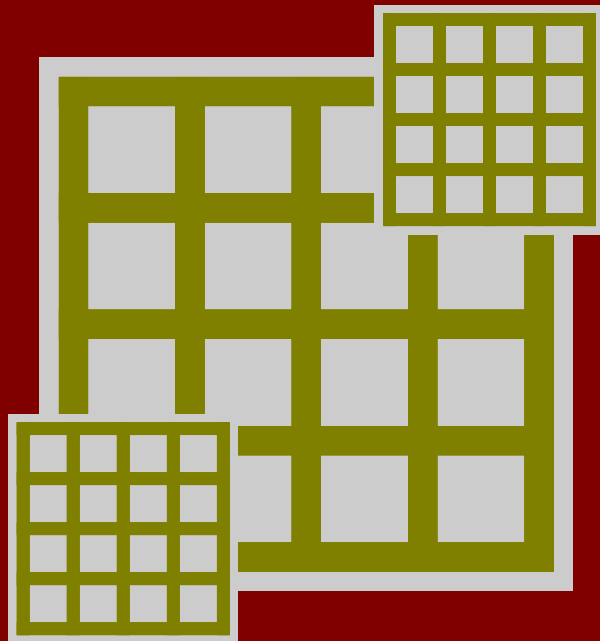
Although installation of T_EX and friends has become relatively easy, fonts always will be a special case. This is a result from the flexibility of T_EX, as well as the fact that T_EX can typeset virtually any language.

The font manual covers the installation of fonts in ConT_EXt and describes in detail how to define typescripts, how to achieve special effects, like hanging punctuation, and how to set up math fonts.

showfont

showunic

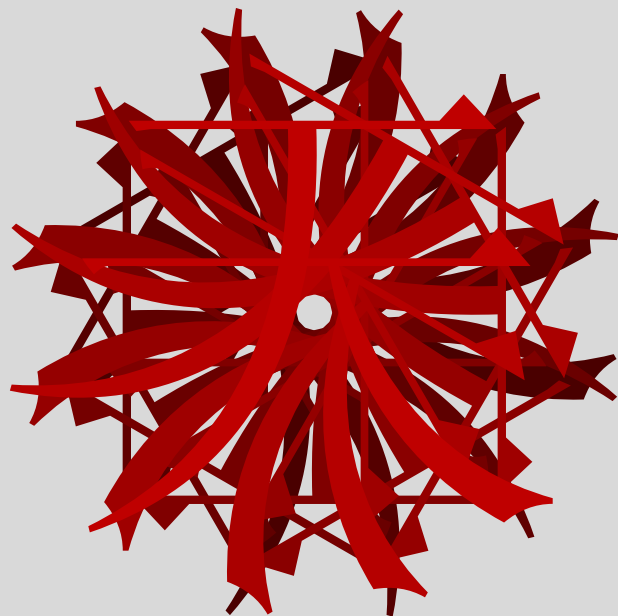
```
name:
documents/general/manuals/mkiv
file:
documents/general/manuals/mkiv
state: unknown
```



T_EXFONT **explained**

Fonts

Installing fonts is one of the nasty parts of using T_EX. This is why ConT_EXt comes with a Perl script called T_EXfont. You can use T_EXfont to generate font metric files in specified encodings, manipulate fonts, creating instances of multiple master fonts, build map files, etc. The script runs on top of afm2t_fm and the mminstance tools.



Chinese in ConT_EXt

Chinese

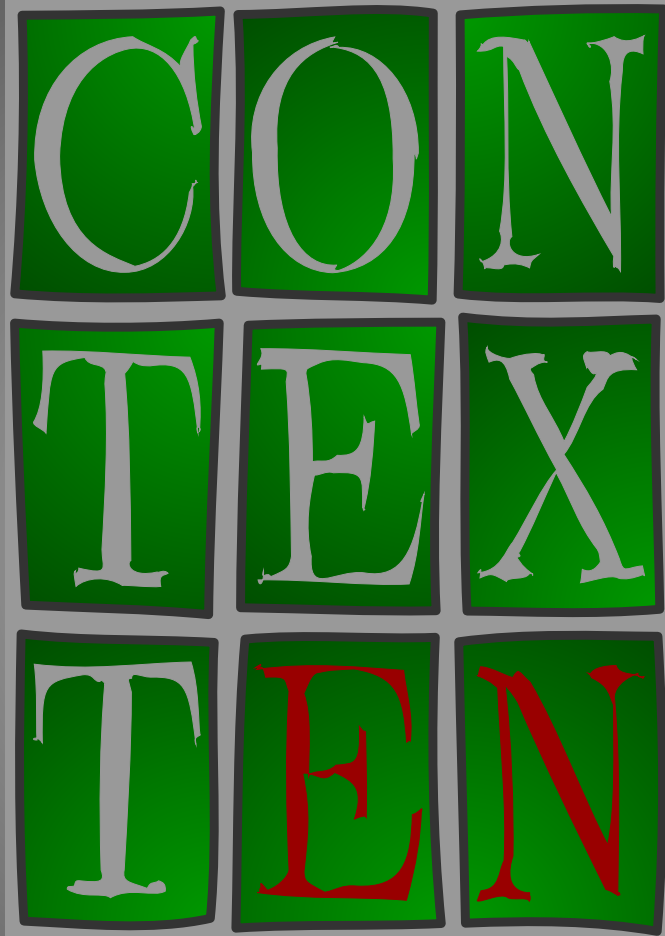
In many aspects, typesetting Chinese differs from typesetting Latin languages. Most noticeably are the pictographic characters, vertical typesetting, multiple numbering systems, and a different way of handling labels.

This manual covers the specific font setups, encoding issues, and mixed Latin and Chinese typesetting.



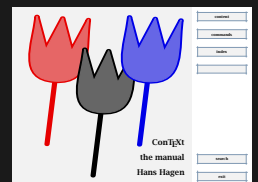
What is ConTeXt

Occasionally I run into a description of ConTeXt that contains observations that are somewhat off. This document provides some insight in why this macro package looks the way it looks. What started out as a T_EX only system evolved via adding MetaPost to the current hybrid system that also uses Lua.



The Manual

This is the big reference manual, the one that is supposed to cover the whole of Con-
T_EXt. This manual is written for MkII but
a lot of it still applies to MkIV. Especially
fonts, encodings and languages are differ-
ent in MkIV. For most commands the user
interface hasn't changed, so don't be fooled
by the fact that this manual is old.



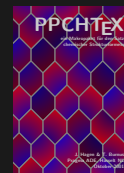
PPCHTEX

a macropackage for typesetting
chemical structure formulas

J. Hagen & A.F. Otten
Pragma ADE, Hasselt NL
November 2001

Chemistry

ppchTeX is a relatively independent macro package that can be used to typeset chemical formulas. These manuals show how it's done. There were also some faq's and a suite with many examples but we removed these.



MATHML

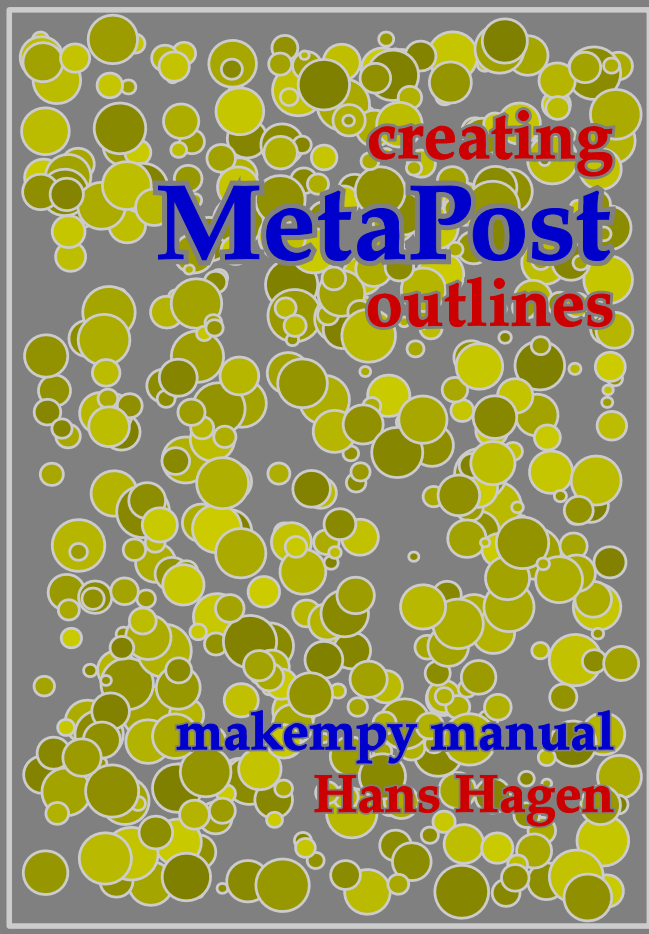
ConT_EXt XML

Pragma ADE / Hasselt NL

MathML support

This (short) manual explains how to invoke MathML support in ConT_EXt. It can be seen as an addendum to the MathML manual.





MetaPost outlines

MakeMPY is a Perl script and some macros that make it possible to create outlines from text typeset by $\text{T}_\text{E}\text{X}$, that can be imported into MetaPost graphics. This toolkit uses pdf $\text{T}_\text{E}\text{X}$, pdftops, pstoeit and Ghostscript, and works with any $\text{T}_\text{E}\text{X}$.

XML in ConT_EXt

introduction
general markup
processing files
defining interfaces
basic workflows
some examples
command reference

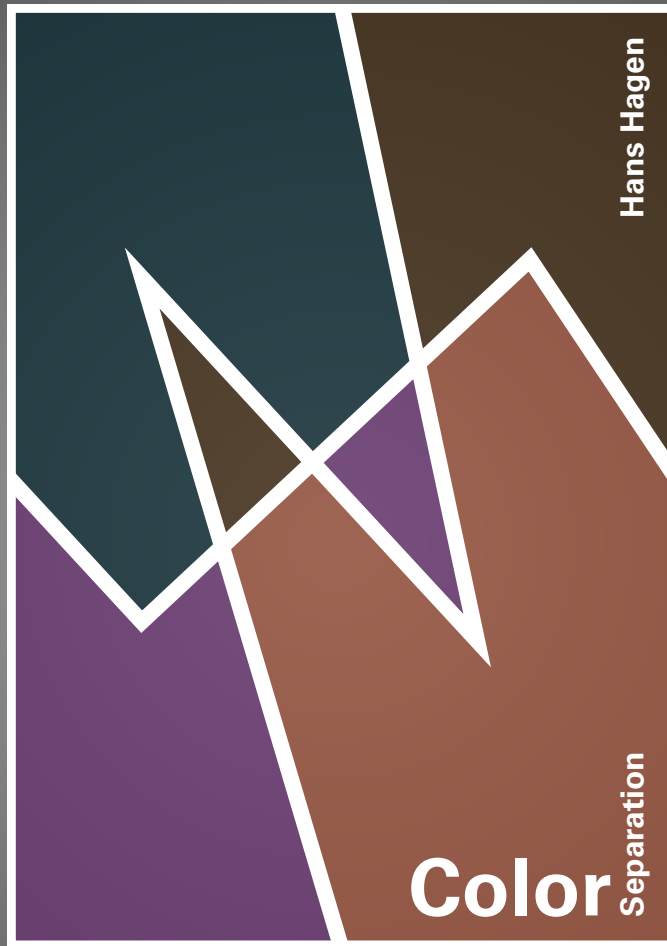
PRAGMA ADE | May 9, 2002

exit begin reference

↩ ↪ ↩ ↪ ↩ ↪ ↩ ↪

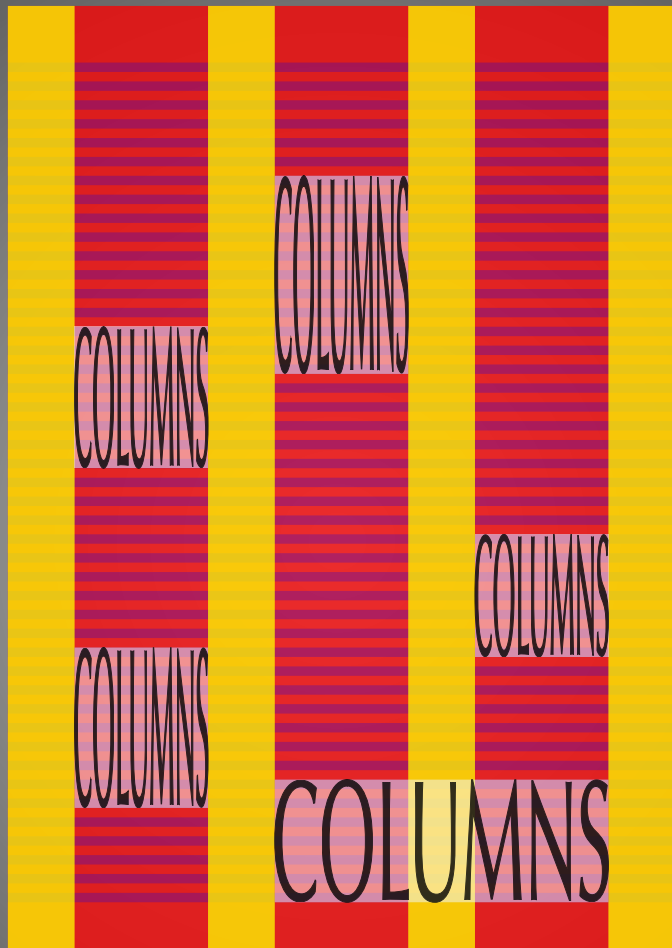
XML

Since T_EX can handle ascii input rather well, it will be no surprise that ConT_EXt can handle xml. In this document we describe the interface to xml. We also provide some examples, tips and tricks. This document is still under construction.



Color separation

This is a manual for those who are forced to deliver their typeset results color separated. The manual describes how to create an instance of a document in a specific color space and channel. Text as well as graphics are covered.



Extreme Columns

Column sets can be used for quite complex but nice looking layouts. (Behind the screens) this mechanism goes to the extremes of what we can do with $\text{\TeX}$'s output routines. With columnsets we try to bridge between sequential makeup and semi automated desk top publishing.



Charts

The flow chart module is an example of combining the power of $\text{T}_{\text{E}}\text{X}$ and MetaPost. You can use this module to define charts in a descriptive way such that parts can be used, and or charts can be combined. The advantage of using this integrated approach (opposite to dedicated programs) is that you have the whole Con $\text{T}_{\text{E}}\text{X}$ t machinery available, like hyperlinks and embedded graphics. Also, by using this module, you have a proper match of fonts between graphics and text.





Weaving PS into PDF

This manual describes the `pstopdf` tool that comes with `ConTEXt`. You can use this tool to convert PostScript images into pdf. The program is actually a wrapper around Ghostscript, but applies some additional trickery and filtering. It also supports watched folders and is suited for interfacing to the `eXaMple` framework.



TEXMFSTART

Hans Hagen – 2003/2006

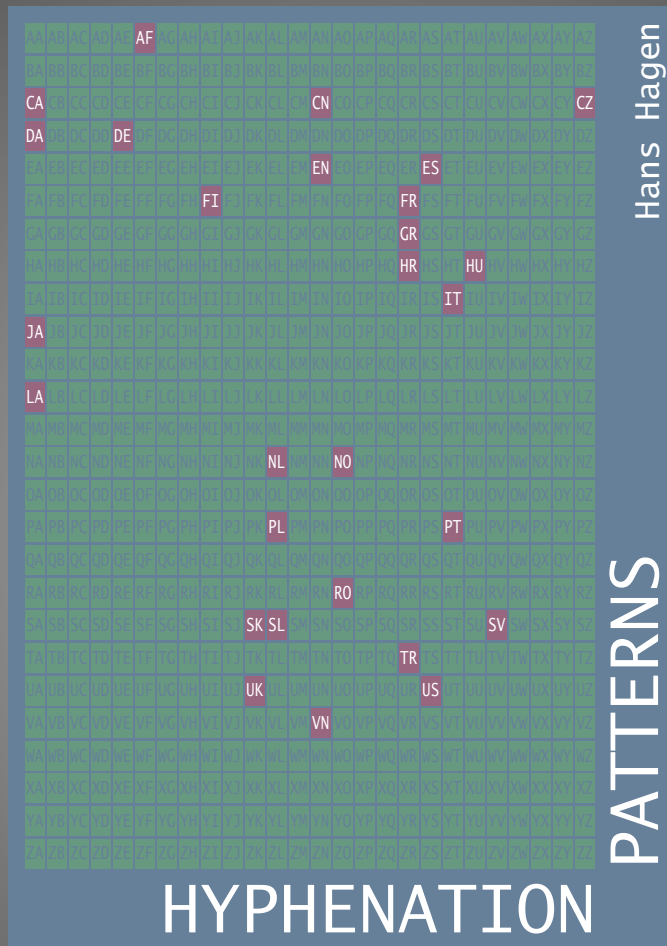
texmfstart & ...tools

This very short manual demonstrates how you can use texmfstart to launch scripts and documents located in your $\text{T}_{\text{E}}\text{X}$ tree. The script uses kpsewhich as well as its own (more aggressive) methods for locating the file.

The $\text{T}_{\text{E}}\text{X}$ tools manual describes a program that actually is a (growing) collection of small utilities that operate on $\text{T}_{\text{E}}\text{X}$ related files and trees.

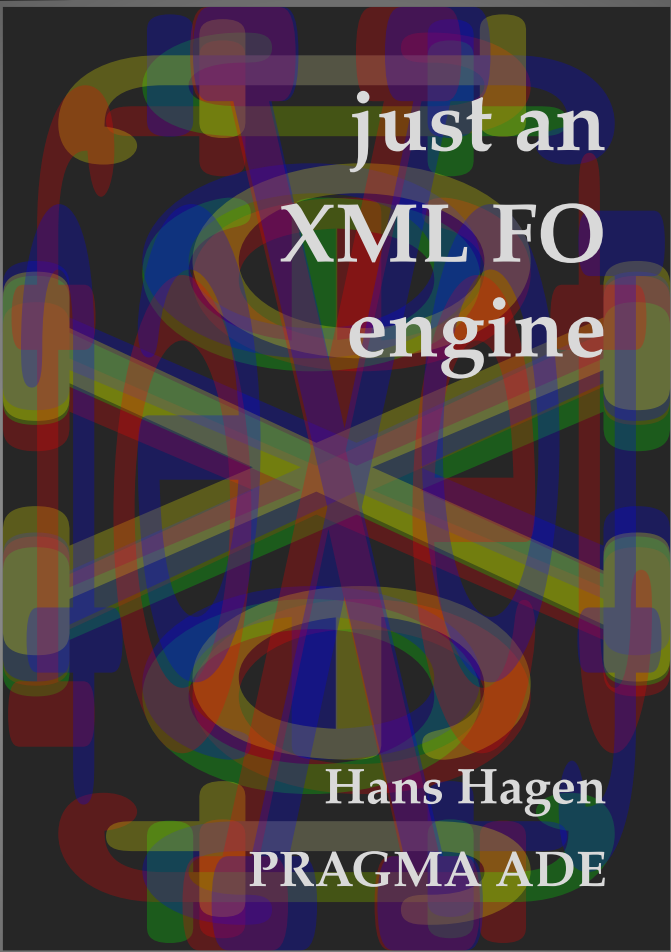
The xmltools manual describes a similar program, this time a collection of utilities that operate on cq. produce xml files and trees.

Finally, the pdftools manual deals with the associated program, that operates on pdf files. This tool is not yet public.



Hyphenation Patterns

Although normally users are not supposed to know the dirty details of pattern management, it may be handy to read this manual at least once, if only to know what to do when for some reason pattern loading fails on your machine. This manual also describes how to apply the cxtxttools program to generate generic pattern files from existing encoding specific files.



just an XML FO engine

Hans Hagen
PRAGMA ADE

foXet

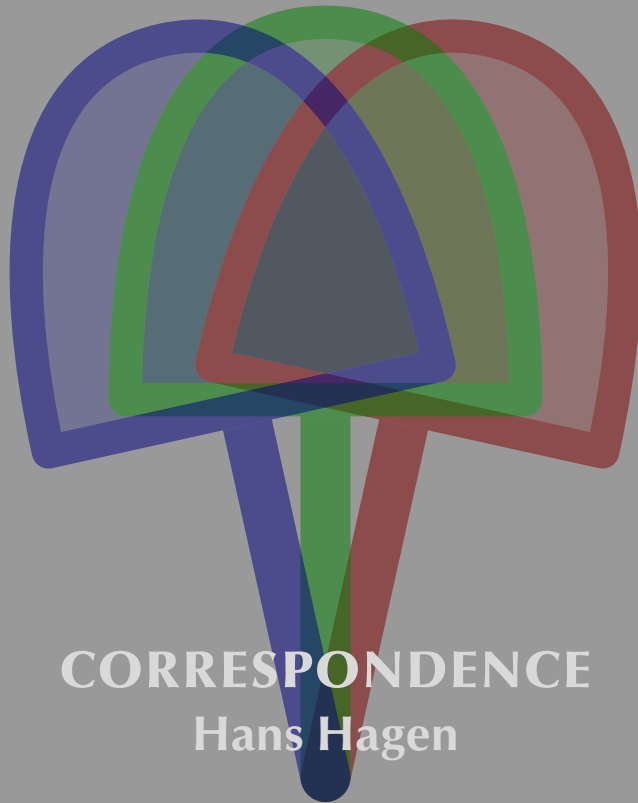
You can see foXet as just another way of processing xml formatting objects. You may use it to process documents coded in (reasonable) xsl-fo or as (textual) graphics format in ConT_EXt documents, a sort of placed xml.



foXet

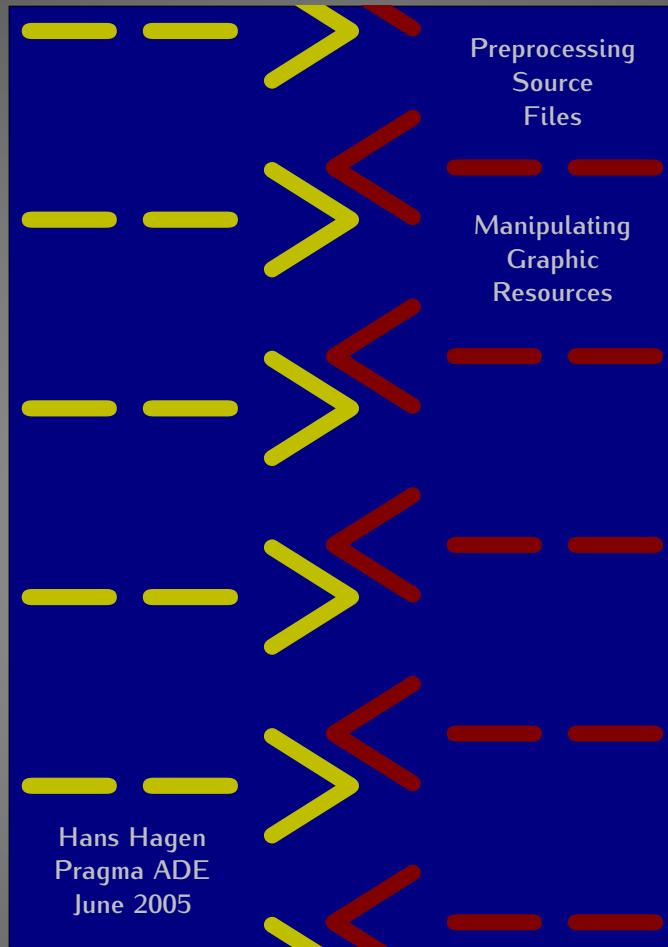
Correspondence

One of the first application at Pragma ADEof $\text{T}_{\text{E}}\text{X}$ was in typesetting letters. Over time the $\text{T}_{\text{E}}\text{X}$ only based system moved to a combination of xml and $\text{T}_{\text{E}}\text{X}$. This manual roughly describes the components that make up such a system. A graphical user interface is provided as well.



Preprocessing and Manipulating

This manual describes the facilities for automatic preprocessing of source files and manipulation of graphics. These features come in handy in automated typesetting workflows and are handled by $\TeX$ exec and rlxtools . The definition files are xml based.

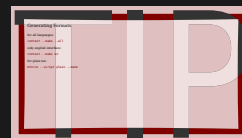


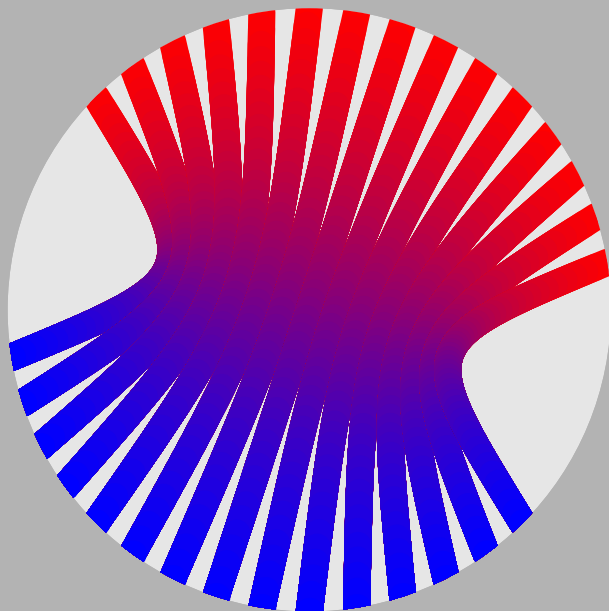


how to install ConT_EXt

Installation

When one uses $\text{teT}_{\text{E}}\text{X}$, $\text{fpT}_{\text{E}}\text{X}$, $\text{gwT}_{\text{E}}\text{X}$, $\text{MikT}_{\text{E}}\text{X}$ or $\text{T}_{\text{E}}\text{X Live}$, installation of ConT_EXt is a breeze. Nevertheless, in this manual, we provide some information on installing ConT_EXt.

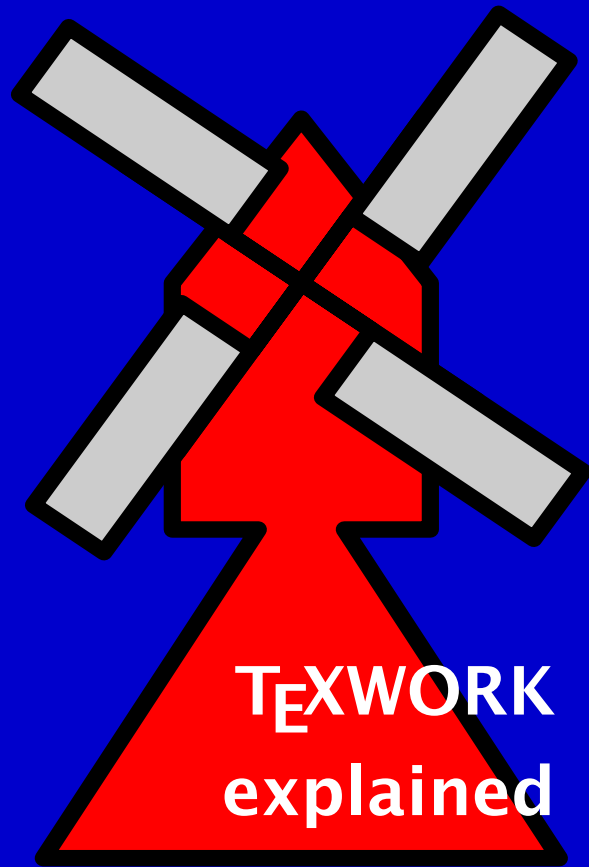




TEXUTIL explained

TEXutil

The TEXutil Perl script deals with files, especially the ConTEXt second pass data file. It moves information around and sorts indexes and lists. This script is the natural companion of TEXexec.

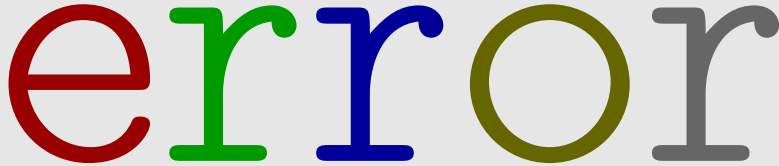


T_EXwork

T_EXwork is our local editing environment. It is a rewrite of the Modula~2 program T_EXedit in Perl/Tk.

Labels

The author of a graphic is not necessarily also its graphic designer. In that case it makes sense to split the design of the graphic elements from the process of adding labels. This document describes how to add text to graphics either or not using the resource (figure) library mechanism.



error



FIGURES

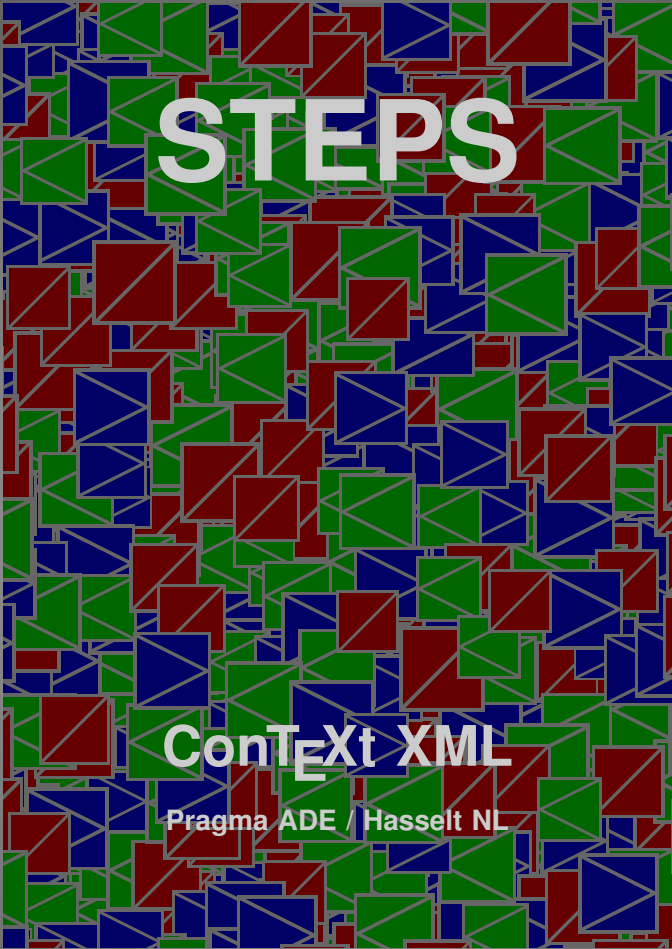
ConT_EXt XML

Pragma ADE / Hasselt NL

Figure Databases

Instead of moving hundreds of graphics around, you can package them in a database. ConT_EXt not only has means to generate such databases, but also can filter the information needed from the corresponding xml files and include graphics by label. Figure bases make it easy to swap high and low resolution graphics.





STEPS

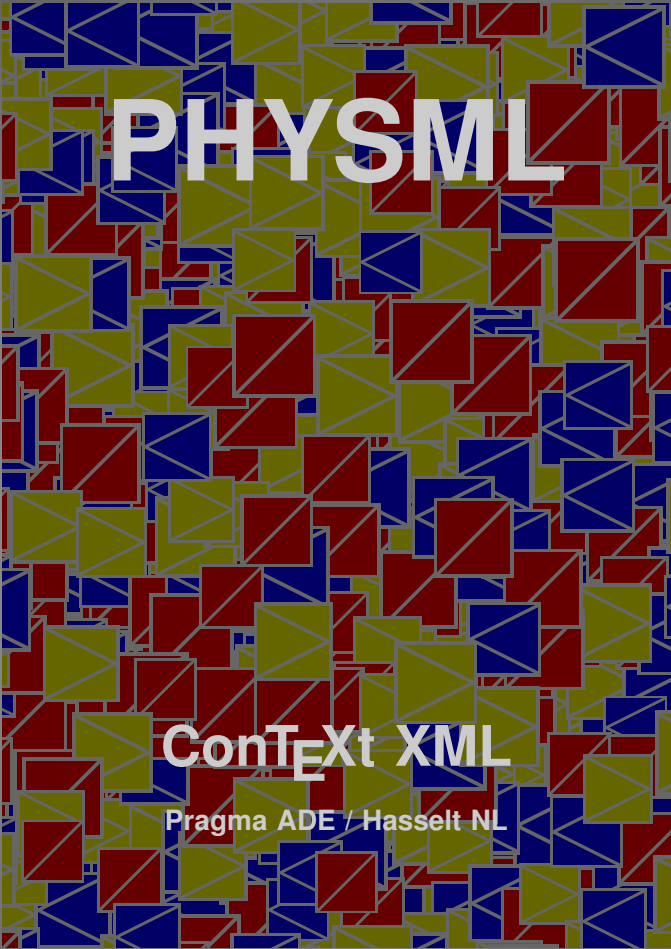
ConT_EXt XML

Pragma ADE / Hasselt NL

Stepcharts

Stepcharts are a specific kind of tabular charts. They are a combination of MetaPost graphics and T_EX code. There is a T_EX as well as xml implementation.





PHYSML

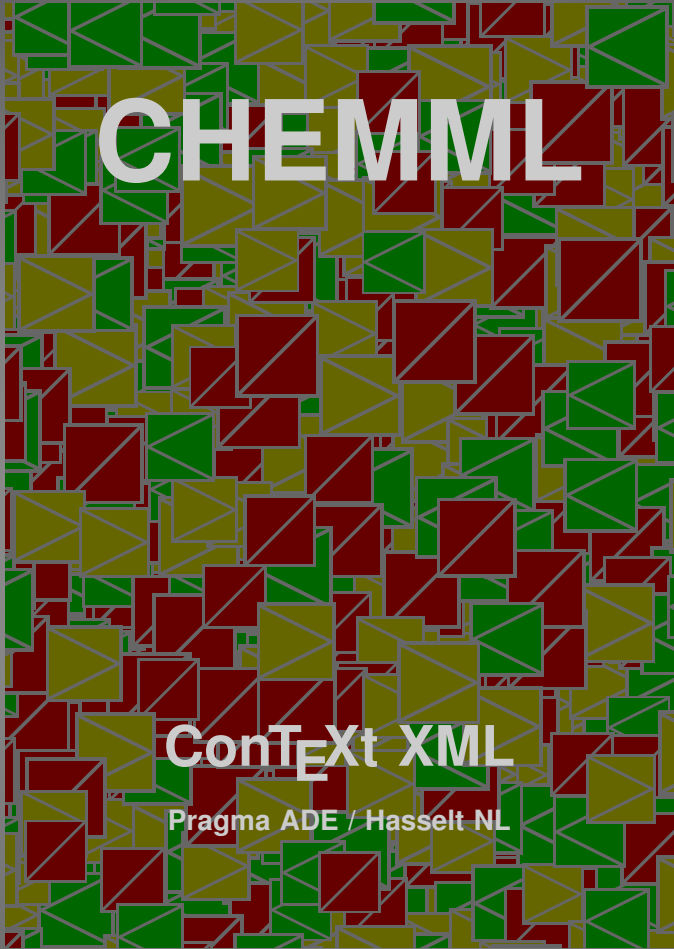
ConT_EXt XML

Pragma ADE / Hasselt NL

PhysML support

Support for physical units is build on top of the MathML engine. The method used is derived from the units module that comes with ConT_EXt.



The image shows the front cover of a book titled 'CHEMML'. The cover has a grey border and a background of a dense, abstract pattern of overlapping squares in green, yellow, and red. The title 'CHEMML' is printed in large, white, bold, sans-serif capital letters at the top. Below it, the subtitle 'Context XML' is printed in a smaller, white, bold, sans-serif font, with the 'T' in 'Context' being smaller and subscripted. At the bottom, the authors 'Pragma ADE / Hasselt NL' are listed in a small, white, sans-serif font.

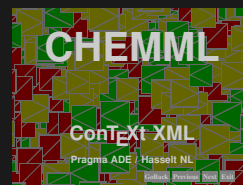
CHEMML

Context XML

Pragma ADE / Hasselt NL

ChemML support

Chemical formulas have their own typographic needs. This module provides a way to code atoms, ions, molecules, and a sequence of reactions.





Widgets

Widgets are interactive elements in (screen based) documents. This manual describes how to use the reference mechanism for advanced hyperlinking, but also discussed how to construct forms. Adding text annotations and page transitions is also discussed. This manual is replaced by the manual on interaction.



XMT DIER

Hans Hagen – Pragma ADE

xmldir

This manual describes how to access information about files on your system from within ConT_EXt. The modules described here are accompanied by features in the T_EXtools script. You can use the styles to generate overviews as well as access properties of files.





Hans Hagen

Modes

Modes

Modes are a convenient way to create styles that server multiple purposes. This manual describes how to enable modes and test for their state. The special system modes that `ConTeXt` sets itself are also explained.